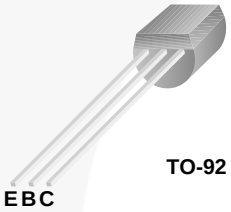
April 2014

2N3906 / MMBT3906 / PZT3906
PNP General-Purpose Amplifier

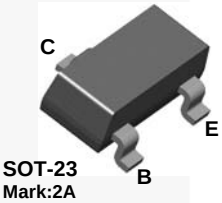
Description

This device is designed for general-purpose amplifier and switching applications at collector currents of 10 mA to 100 mA.

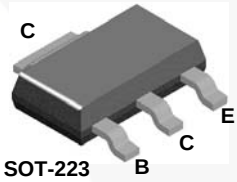
2N3906



MMBT3906



PZT3906



Ordering Information

Part Number	Marking	Package	Packing Method	Pack Quantity
2N3906BU	2N3906	TO-92 3L	Bulk	10000
2N3906TA	2N3906	TO-92 3L	Ammo	2000
2N3906TAR	2N3906	TO-92 3L	Ammo	2000
2N3906TF	2N3906	TO-92 3L	Tape and Reel	2000
2N3906TFR	2N3906	TO-92 3L	Tape and Reel	2000
MMBT3906	2A	SOT-23 3L	Tape and Reel	3000
PZT3906	3906	SOT-223 4L	Tape and Reel	2500

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Absolute Maximum Ratings[1]

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
V_{CEO}	Collector-Emitter Voltage	-40	V
V_{CBO}	Collector-Base Voltage	-40	V
V_{EBO}	Emitter-Base Voltage	-5.0	V
I_C	Collector Current - Continuous	-200	mA
T_J, T_{STG}	Operating and Storage Junction Temperature Range	-55 to +150	°C

Note:

1. These ratings are based on a maximum junction temperature of 150°C.
   These are steady-state limits. Fairchild Semiconductor should be consulted on applications involving pulsed or low-duty cycle operations.

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Maximum			Unit
		2N3906[3]	MMBT3906[2]	PZT3906[3]	
P_D	Total Device Dissipation	625	350	1,000	mW
	Derate Above 25°C	5.0	2.8	8.0	mW/°C
$R_{\theta JC}$	Thermal Resistance, Junction to Case	83.3			°C/W
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	200	357	125	°C/W

Notes:

2. Device is mounted on FR-4 PCB 1.6 inch X 1.6 inch X 0.06 inch.
3. PCB size: FR-4, 76 mm x 114 mm x 1.57 mm (3.0 inch x 4.5 inch x 0.062 inch) with minimum land pattern size.

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Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Min.	Max.	Unit
OFF CHARACTERISTICS					
$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage[4]	I_C = -1.0 mA, I_B = 0	-40		V
$V_{(BR)CBO}$	Collector-Base Breakdown Voltage	I_C = -10 µA, I_E = 0	-40		V
$V_{(BR)EBO}$	Emitter-Base Breakdown Voltage	I_E = -10 µA, I_C = 0	-5.0		V
I_{BL}	Base Cut-Off Current	V_{CE} = -30 V, V_{BE} = 3.0 V		-50	nA
I_{CEX}	Collector Cut-Off Current	V_{CE} = -30 V, V_{BE} = 3.0 V		-50	nA
ON CHARACTERISTICS					
h_{FE}	DC Current Gain[4]	I_C = -0.1 mA, V_{CE} = -1.0 V	60		
		I_C = -1.0 mA, V_{CE} = -1.0 V	80		
		I_C = -10 mA, V_{CE} = -1.0 V	100	300	
		I_C = -50 mA, V_{CE} = -1.0 V	60		
		I_C = -100 mA, V_{CE} = -1.0V	30		
V_{CE}(sat)	Collector-Emitter Saturation Voltage	I_C = -10 mA, I_B = -1.0 mA		-0.25	V
		I_C = -50 mA, I_B = -5.0 mA		-0.40	
V_{BE}(sat)	Base-Emitter Saturation Voltage	I_C = -10 mA, I_B = -1.0 mA	-0.65	-0.85	V
		I_C = -50 mA, I_B = -5.0 mA		-0.95	
SMALL SIGNAL CHARACTERISTICS					
f_T	Current Gain - Bandwidth Product	I_C = -10 mA, V_{CE} = -20 V, f = 100 MHz	250		MHz
C_{obo}	Output Capacitance	V_{CB} = -5.0 V, I_E = 0, f = 100 kHz		4.5	pF
C_{ibo}	Input Capacitance	V_{EB} = -0.5 V, I_C = 0, f = 100 kHz		10.0	pF
NF	Noise Figure	I_C = -100 µA, V_{CE} = -5.0 V, R_S = 1.0 kΩ, f = 10 Hz to 15.7 kHz		4.0	dB
SWITCHING CHARACTERISTICS					
t_d	Delay Time	V_{CC} = -3.0 V, V_{BE} = -0.5 V, I_C = -10 mA, I_{B1} = -1.0 mA		35	ns
t_r	Rise Time			35	ns
t_s	Storage Time	V_{CC} = -3.0 V, I_C = -10 mA, I_{B1} = I_{B2} = -1.0 mA		225	ns
t_f	Fall Time			75	ns

Note:

4. Pulse test: pulse width ≤ 300 µs, duty cycle ≤ 2.0%.

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Typical Performance Characteristics

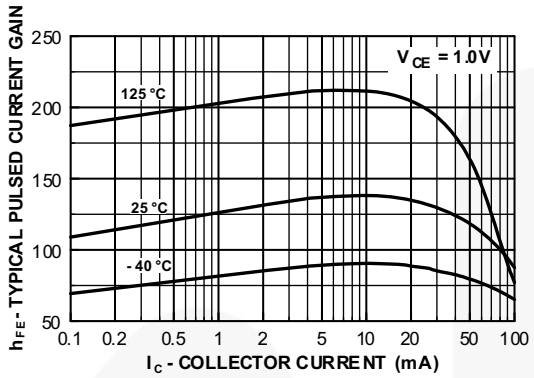


Figure 1. Typical Pulsed Current Gain vs. Collector Current

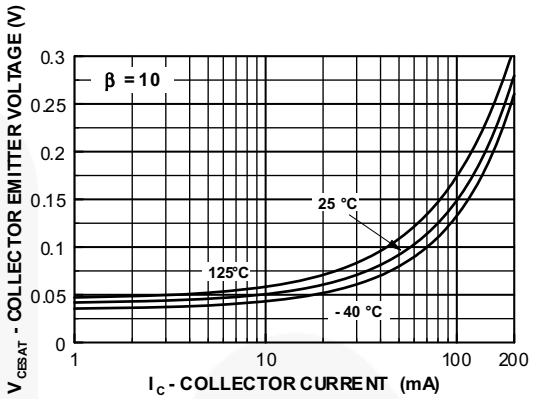


Figure 2. Collector-Emitter Saturation Voltage vs. Collector Current

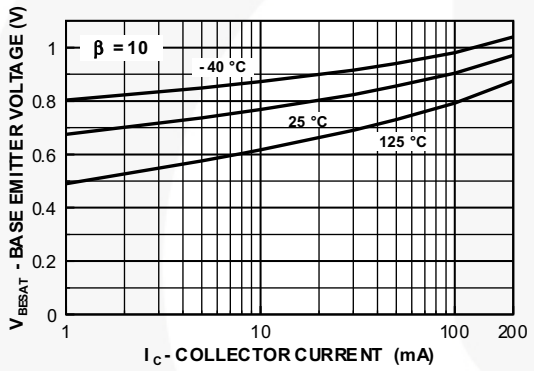


Figure 3. Base-Emitter Saturation Voltage vs. Collector Current

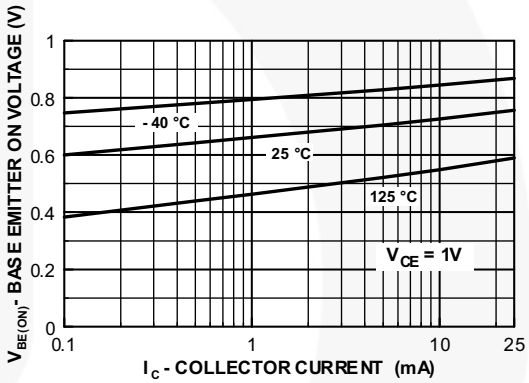


Figure 4. Base-Emitter On Voltage vs. Collector Current

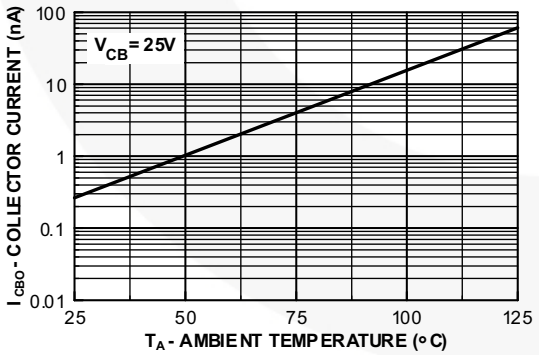


Figure 5. Collector Cut-Off Current vs. Ambient Temperature

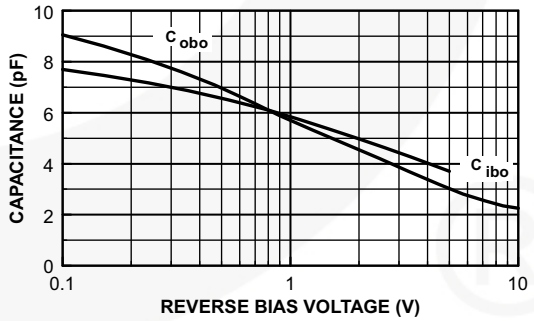


Figure 6. Common-Base Open Circuit Input and Output Capacitance vs. Reverse Bias Voltage

Typical Performance Characteristics (Continued)

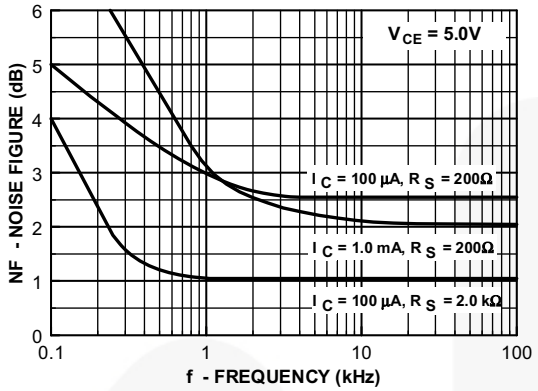


Figure 7. Noise Figure vs. Frequency

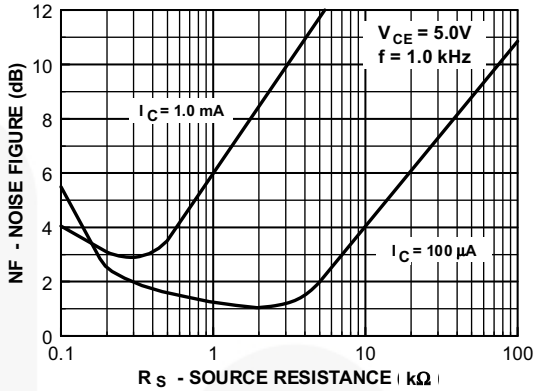


Figure 8. Noise Figure vs. Source Resistance

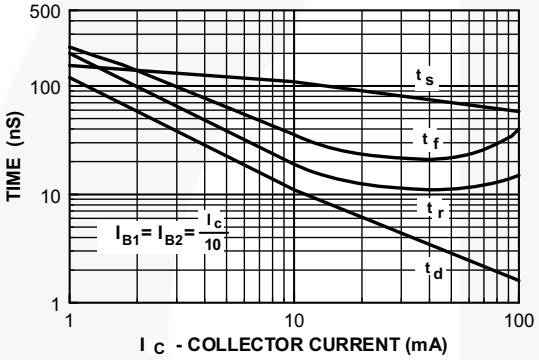


Figure 9. Switching Times vs. Collector Current

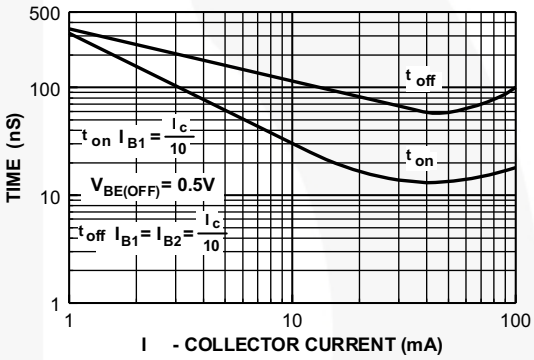


Figure 10. Turn-On and Turn-Off Times vs. Collector Current

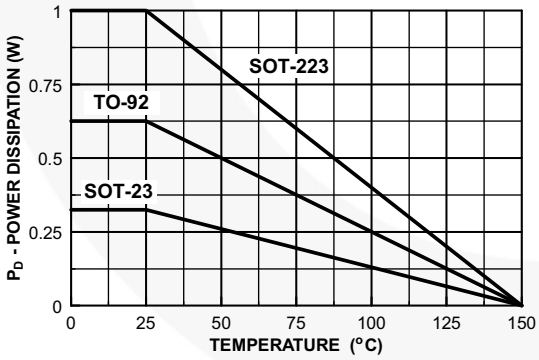


Figure 11. Power Dissipation vs. Ambient Temperature

Typical Performance Characteristics (Continued)

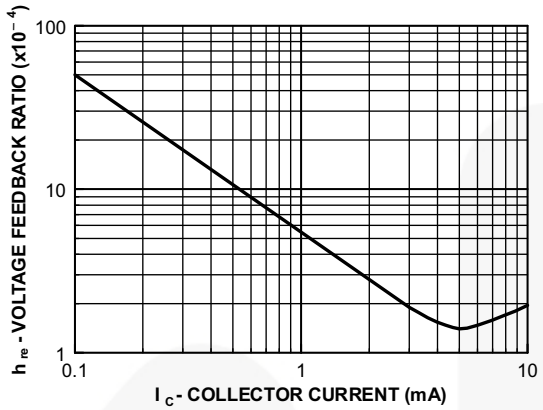


Figure 12. Voltage Feedback Ratio

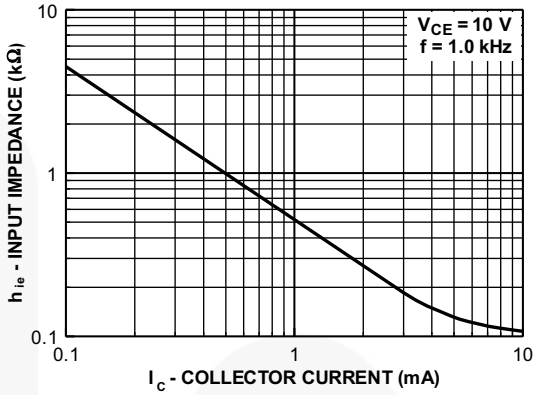


Figure 13. Input Impedance

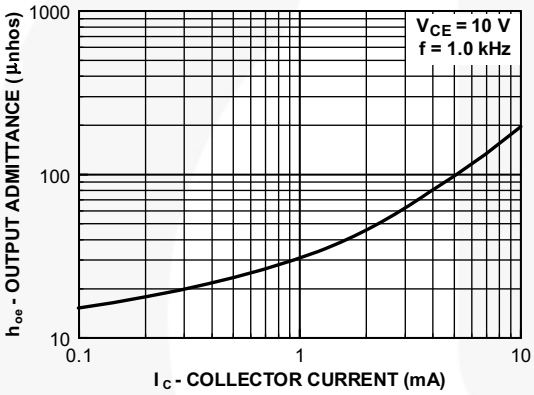


Figure 14. Output Admittance

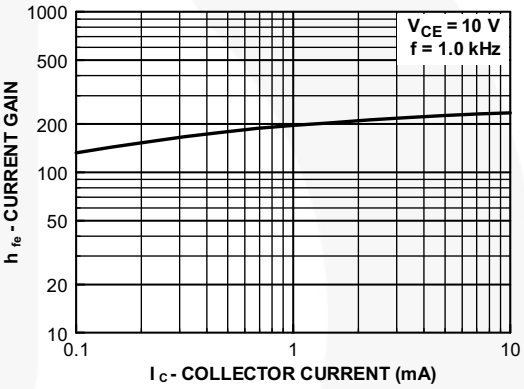
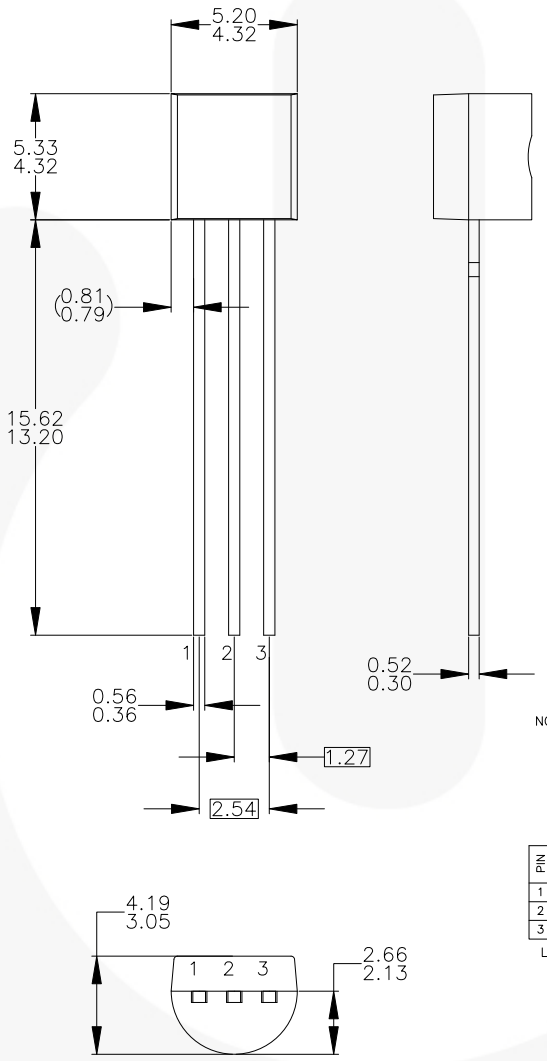


Figure 15. Current Gain

Physical Dimensions

TO-92 (Bulk)



NOTES: UNLESS OTHERWISE SPECIFIED

A) DRAWING WITH REFERENCE TO JEDEC TO-92 RECOMMENDATIONS.
B) ALL DIMENSIONS ARE IN MILLIMETERS.
C) DRAWING CONFORMS TO ASME Y14.5M-1994.
D) TO-92 (92,94,96,97,98) PIN CONFIGURATION:

PIN	92			94			96			97			98		
	P	F	M	P	F	M	B	F	M	P	F	M	P	F	M
1	E	S	S	E	S	S	B	D	G	C	G	D	C	G	D
2	B	D	G	C	G	D	E	S	S	B	D	G	E	S	S
3	C	G	D	B	D	G	C	G	D	E	S	S	B	D	G

LEGEND:
P – BIPOLAR
F – JFET
M – DMOS
E – EMITTER
B – BASE
C – COLLECTOR
D – DRAIN
S – SOURCE
G – GATE

E) FOR PACKAGE 92, 94, 96, 97 AND 98: PIN CONFIGURATION DRAIN "D" AND SOURCE "S" ARE INTERCHANGEAGLE AT JFET "F" OPTION.
F) DRAWING FILENAME: MKT-ZA03DREV3.

Figure 16. 3-LEAD, TO92, JEDEC TO-92 COMPLIANT STRAIGHT LEAD CONFIGURATION (OLD TO92AM3)

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Physical Dimensions (Continued)

TO-92 (Ammo, Tape and Reel)

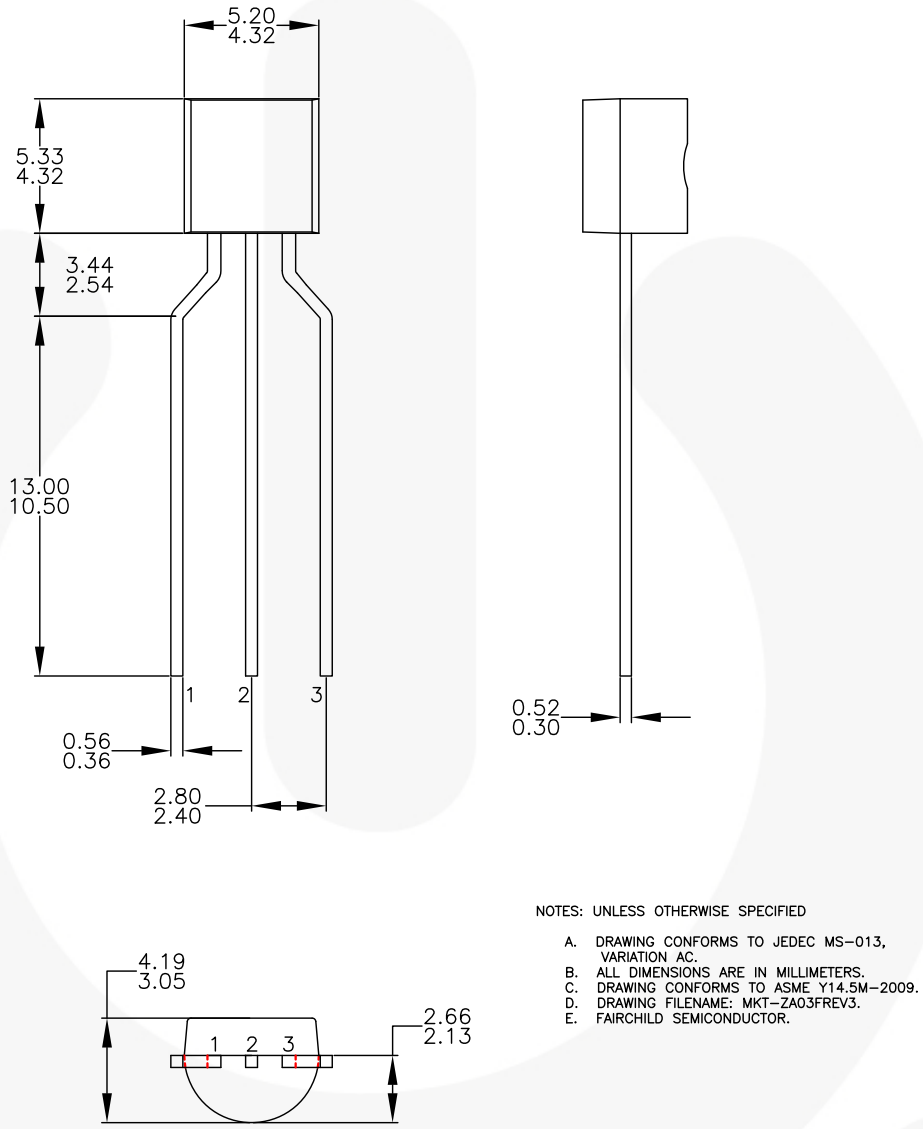

Figure 17. 3-LEAD, TO92, MOLDED 0.200 IN LINE SPACING LEAD FORM (J61Z OPTION)

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Physical Dimensions (Continued)

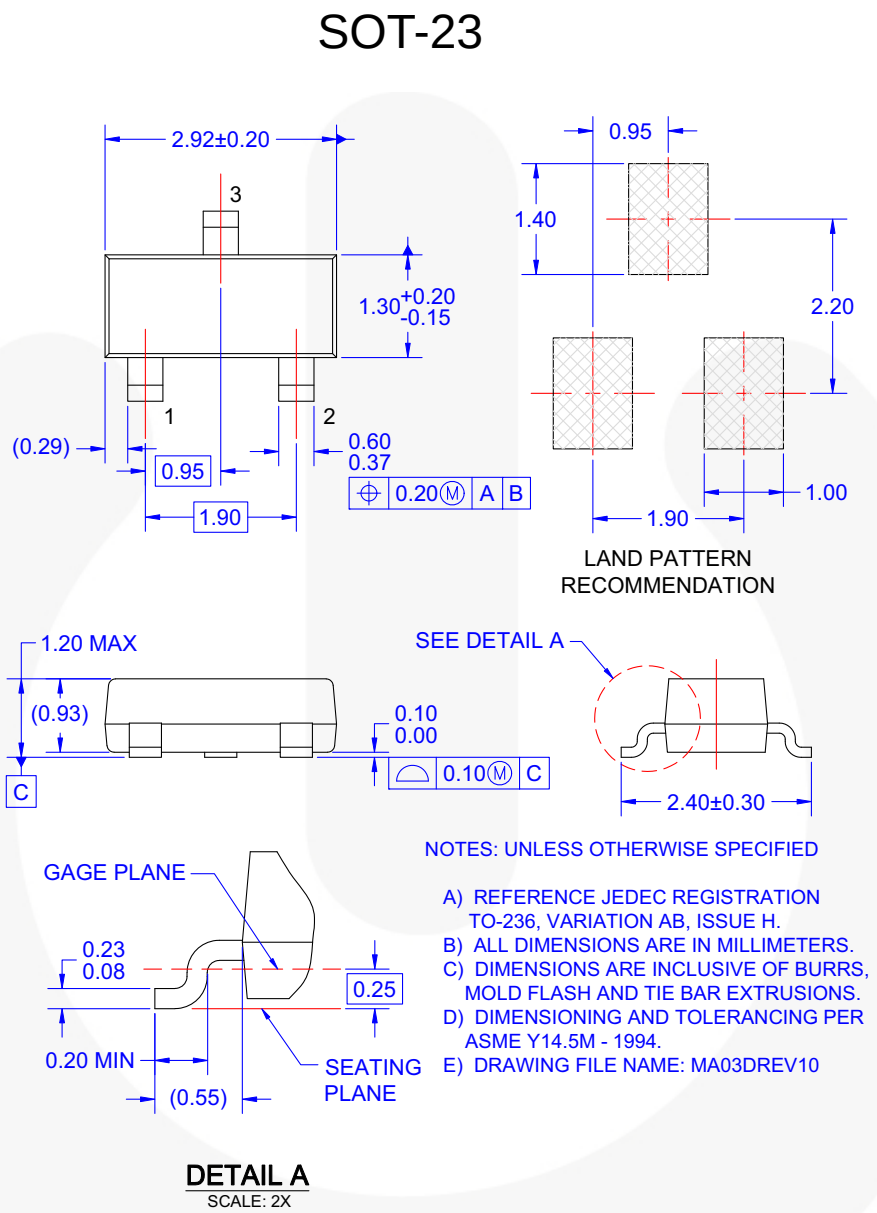

Figure 18. 3-LEAD, SOT23, JEDEC TO-236, LOW PROFILE

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Physical Dimensions (Continued)

SOT-223 4L

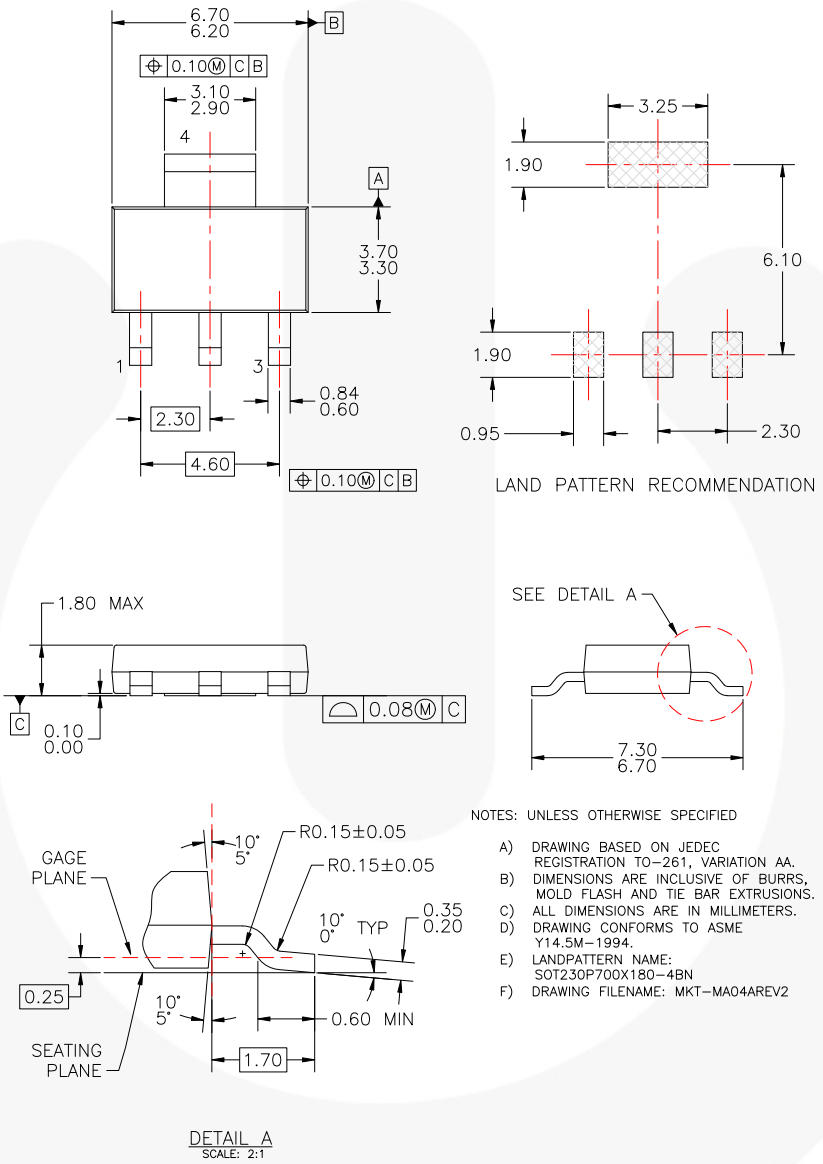

Figure 19. MOLDED PACKAGE, SOT-223, 4-LEAD

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Advance Information	Formative / In Design	Datasheet contains the design specifications for product development. Specifications may change in any manner without notice.
Preliminary	First Production	Datasheet contains preliminary data; supplementary data will be published at a later date. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve design.
No Identification Needed	Full Production	Datasheet contains final specifications. Fairchild Semiconductor reserves the right to make changes at any time without notice to improve the design.
Obsolete	Not In Production	Datasheet contains specifications on a product that is discontinued by Fairchild Semiconductor. The datasheet is for reference information only.

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